

Advancements and implications of semantic reconstruction of continuous language from non-invasive brain recordings

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Semantic reconstruction of continuous language from non-invasive brain recordings is an emerging research field that aims to decode the meaning of words, sentences,^[1] or even entire narratives from neural activity patterns recorded using non-invasive techniques like electroencephalography or magnetoencephalography.^[2] Semantic reconstruction of continuous language from non-invasive brain recordings can potentially transform our understanding of how the brain processes language.

Tang et al.^[3] presented a novel method for reconstructing continuous language from cortical semantic representations of functional magnetic resonance imaging (fMRI) recording of neural activity in the brains of three human participants while they listened to spoken stories. They decoded the fMRI signals using a neural network and reconstructed the auditory and semantic content of the stories. Their findings are crucial in developing brain–computer interfaces (BCIs) that can facilitate communication between humans and machines. Their research developed a BCI that can decode continuous language from non-invasive recordings to construct cortical semantic representations and reconstruct word sequences that recover the meaning of perceived speech, imagined speech, and even silent videos. Their study explored the viability of non-invasive language BCIs, which

may provide advice or references for potential scientific and practical applications in the future.

Tang et al.'s method introduces an innovative approach to explore language processing in the brain with fMRI. While their approach does not surmount fMRI's inherent low temporal resolution of fMRI, it employs a strategy that generates candidate word sequences, helping to gathering insights into the neural substrates and mechanisms associated with language processing. This method offers a nuanced perspective by leveraging some aspects of the fMRI data and grounding its analysis on certain assumptions about the statistical patterns in natural language processing. Conventional fMRI studies have grappled with challenges when delving into language processing due to the inherent lag in the blood oxygen level-dependent response. While not real-time, Tang et al.'s method, offers a direction that deviates from traditional static maps, like those presented by Huth et al.,^[4] and prompts considerations into a richer understanding of the brain's approach to language.

BCIs have been instrumental in restoring communication capabilities to individuals who have lost the ability to speak. Previously, these technologies primarily relied on invasive methods, which were impractical for broader applications. The technological novelty of this BCI lies in its ability to

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decode continuous language from cortical semantic representations. Historically, fMRI's low temporal resolution posed a significant hurdle to achieving this feat. The authors tackled this challenge through an ingenious approach by generating candidate word sequences and scoring the likelihood of each candidate evoking the recorded brain responses. They accomplished this by employing an encoding model that predicts the subject's brain responses to natural language.

Furthermore, the authors demonstrated the BCI's versatility by showing that it could decode language from multiple regions across the cortex. Another remarkable aspect is the emphasis on mental privacy, with the study reporting that successful decoding requires subject cooperation. As this technology becomes more advanced, its implementation of such technology also raises ethical considerations, particularly regarding mental privacy and the potential for misuse. Developing appropriate guidelines and regulations to protect individuals' privacy is vital. Another significant ethical concern is informed consent. Individuals who participate in studies involving non-invasive brain recordings should be fully informed of the risks and benefits of the study and should provide informed consent before participating.

One of the key future directions of this field is developing more accurate and efficient decoding algorithms. While the current decoding algorithms have shown promising results, there is still room for improvement. Future research should focus on developing algorithms that are more robust to individual differences and can decode language in real-time.^[5] Another important future direction is exploring the neural mechanisms underlying language processing. While we have made significant progress in decoding language from non-invasive brain recordings, our understanding of the neural mechanisms underlying language processing remains limited. Future research should focus on elucidating these mechanisms to improve our ability to decode language from brain recordings. Another important future direction is translating this technology into clinical settings. Therefore, future research should focus on developing clinical applications of this technology and evaluating its efficacy in clinical settings.

Overall, while semantic reconstruction of continuous language from non-invasive brain recordings is a promising technology with many potential applications, there are still significant technical and ethical challenges remain that must be addressed. By continuing to push the boundaries of this technology while adhering to ethical principles and ensuring regulatory oversight and transparency, we can maximize its benefits while minimizing its risks.

AUTHOR CONTRIBUTIONS

Zhao Chen, Yanping Wang and Nannan Shi conceived and developed this commentary. **Zhao Chen:** Writing—original draft. **Ning Liang, Haili Zhang, Huizhen Li, and Xiangwei Dai** edited and approved the final version.

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CONFLICT OF INTEREST STATEMENT

All authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

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REFERENCES

1. Mitchell TM, Shinkareva SV, Carlson A, et al. Predicting human brain activity associated with the meanings of nouns. *Science*. 2008; 320(5880):1191-1195. <https://doi.org/10.1126/science.1152876>
2. Haynes J.-D, Rees G. Decoding mental states from brain activity in humans. *Nat Rev Neurosci*. 2006;7(7):523-534. <https://doi.org/10.1038/nrn1931>
3. Tang J, LeBel A, Jain S, Huth AG. Semantic reconstruction of continuous language from non-invasive brain recordings. *Nat Neurosci*. 2023;26(5):858-866. <https://doi.org/10.1038/s41593-023-01304-9>
4. Huth AG, De Heer WA, Griffiths TL, Theunissen FE, Gallant JL. Natural speech reveals the semantic maps that tile human cerebral cortex. *Nature*. 2016;532(7600):453-458. <https://doi.org/10.1038/nature17637>
5. Wang B, Chen Y, Chen K, Lu H, Zhang Z. From local properties to brain-wide organization: a review of intraregional temporal features in functional magnetic resonance imaging data. *Hum Brain Mapp*. 2023; 44(9):3926-3938. <https://doi.org/10.1002/hbm.26302>

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